

**Awarding Body:** ARDEN UNIVERSITY

**Programme Name:** BSc in Computing with Foundation Year

**Module Name (and Part if applicable):** Distributed and Cloud Computing

**Assessment Title:** Project

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**Word Count:** 3866

Please refer to the Word Count Policy on your Module Page for guidance

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## Introduction

Organisations now design and operate digital systems through cloud computing because it offers scalable on-demand computing resources, which eliminate the requirement for physical infrastructure ownership. Small and medium-sized businesses like Flynet Transportation can achieve better flexibility, cost efficiency, and operational continuity through cloud-based distributed solutions when they move away from manual dispatch service management. The project outlines a complete infrastructure design which explains how to implement Flynet's fleet-management system on Microsoft Azure through IaaS and PaaS components within a Public Cloud deployment model.

The system design fulfils Flynet operational requirements through distributed systems principles, which deliver remote dispatch management and real-time driver tracking and data processing while ensuring scalability and reliability, and maintaining security standards. The infrastructure combines Azure services through private networking to create a resilient, modular system which includes Virtual Machine Scale Sets (VMSS), Load Balancer, PostgreSQL Flexible Server, and Blob Storage. The report includes detailed analytical explanations for design decisions, together with scalability and reliability assessments, cost projections, and strategic guidance for upcoming enhancements.

## **Task 1**

### **Infrastructure Design**

#### Context and Aim

Flynet Transportation Limited is a small private taxi company operating within a five-mile radius of a local train station. The dispatcher operates through a manual process of recording journey requests by hand before using radio communication to contact drivers. This process depends on the dispatcher's personal knowledge, and there is no centralised digital record. The manual coordination system creates operational inefficiencies while making data observation difficult and prevents the organisation from expanding its operations when demand is high (Valacich, George, and Hoffer, 2023).

The project establishes a protected cloud-based system that enables the open-source web application to transform Flynet's fleet management operations. The proposed architecture implements IaaS virtual machines from Microsoft Azure and PaaS database and storage components. The cloud platform supports remote dispatch operations, which removes the requirement for physical offices while allowing businesses to grow step by step into new service areas. The organisation chose cloud-based technology because Laudon and Laudon (2022) explain that modern businesses select cloud-based solutions because they offer operational flexibility and cost-effectiveness.

The system design follows distributed computing principles through its ability to scale and maintain fault tolerance while keeping application logic independent from data storage. The web application and dispatcher interface are deployed in a web tier that scales horizontally via a virtual machine scale set, while booking and fleet data are stored in a managed relational database service. The multi-layered system design provides dependable service operations for different workload conditions (Chandrasekaran, 2015).

Organisations can reach operational resilience and security benefits through network segmentation with controlled access policies, encrypted communication, and continuous monitoring systems. The system operates under the shared responsibility model of cloud computing because it distributes platform management duties between the cloud provider and access control and data processing responsibilities between the organisation (Jansen and Grance, 2011).

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## Research on Typical System Requirements

Typical fleet-management platforms for taxi and transport services consist of three main components: a dispatcher interface for booking management, a driver interface for status reporting and navigation, and a backend service for storing trip data, vehicle details, and driver availability information. The systems require immediate communication, joint database access, and protected internet connectivity (Traccar, 2025).

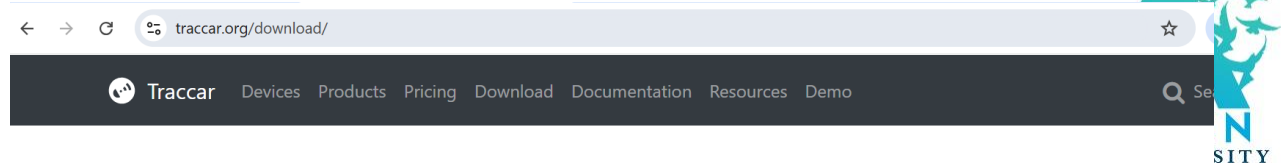
For this project, the selected open-source software is the Traccar GPS Tracking System, available free of charge at:

<https://www.traccar.org> (Accessed 10 November 2025).

Traccar is a widely used fleet-management and GPS-tracking platform written in Java, using the Netty asynchronous framework for network communication and a relational database (PostgreSQL, MySQL, or H2) for data storage. The system provides dispatchers and administrators with web-based access, while drivers use mobile applications, and the system enables real-time data exchange through REST / WebSocket APIs. The system provides core features which support vehicle and driver management, geofencing, route history, and live map tracking to meet Flynet's requirements for modern cloud-based dispatching.

The official download page provides installation packages for Linux and Windows, as well as Docker images, as shown in Figure 1. The open-source nature of Traccar becomes evident in this demonstration, which shows its readiness for cloud deployment with no licensing barriers.





## Download Server

The current official release of the Traccar GPS tracking system. All packages include the server and the web app.

| Operating System                                                                                      | Link                                          |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------|
|  Docker Image        | <a href="#">information</a>                   |
|  Linux x64           | <a href="#">traccar-linux-64-6.10.0.zip</a>   |
|  Linux ARM (AArch64) | <a href="#">traccar-linux-arm-6.10.0.zip</a>  |
|  Windows x64         | <a href="#">traccar-windows-64-6.10.0.zip</a> |
|  Other               | <a href="#">information</a>                   |

If you are looking for a specific older version, please check [older versions](#).

Traccar software is completely free and open source. No limitations on commercial or private use.

*Figure 1: The Traccar open-source fleet-management software download page (Traccar, 2025).*

Because Traccar natively supports PostgreSQL, it integrates seamlessly with the Azure Database for PostgreSQL used in this cloud design. The system employs this method to preserve compatibility and data consistency while delivering high performance in distributed systems.

The application architecture of open-source solutions Traccar and FleetBase uses a layered design, which divides the application tier, consisting of web servers and APIs, from the data tier that includes databases and storage systems to achieve better scalability, maintainability, and fault isolation (Chandrasekaran, 2015). Based on these principles, the proposed Azure architecture for Flynet isolates the web tier, hosting the dispatcher and driver interfaces, from the data tier, which stores booking and operational data in a managed PostgreSQL service. The system design enables Flynet to execute operational updates and track driver availability in real time, and provides dispatchers working from home with secure remote access.

## Overall Architecture Design

The cloud infrastructure supports the open-source fleet-management application Traccar, which enables real-time driver tracking, booking management, and dispatch coordination through a web interface and mobile client connections (Traccar, 2025). The web server-based Traccar system uses relational database technology to store journey, vehicle, and driver information, which enables deployment in cloud-based environments.

The solution exists in Microsoft Azure through a specific Resource Group (FlynetFleet-RG), which enables cost management and access control. A Virtual Network (vnet-flynet) provides an isolated networking environment and is segmented into two subnets: a web tier (snet-web) that runs the application servers and a data tier (snet-data) that hosts the relational database and storage services. The system design includes separate tiers which allow individual system development while protecting against attacker system-to-system movement after initial system penetration. The system design pattern demonstrates the separation of application logic from data storage, which follows the distributed system design principles described by Chandrasekaran (2015).

Traccar is deployed on a Virtual Machine Scale Set to allow the number of running server instances to increase or decrease based on system demand. The system operates at a consistent level during peak hours because of this design which also minimizes expenses when the system operates at lower capacity. The PostgreSQL database selection in the data tier maintains full compatibility with Traccar application components while eliminating the requirement for proprietary licensing expenses.

The architecture design provides remote dispatcher access, dependable data management, and flexible service delivery, which meets Flynet's operational requirements.

## Overall Cloud Infrastructure Design

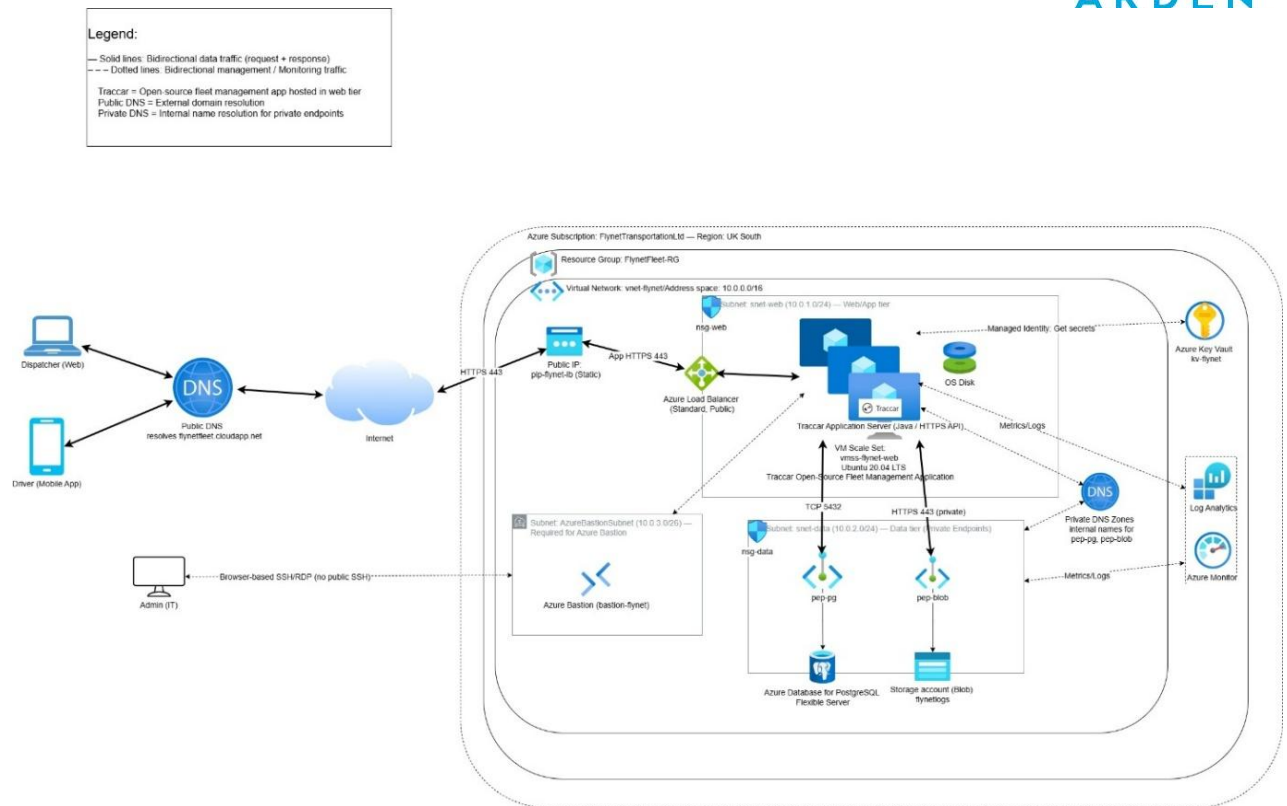


Figure 2: The overall architecture diagram with the resources included.



Infrastructure Components and Configuration

Virtual Network and Subnets

| Component                     | Configuration                 | Purpose                                                                                                                                                                           |
|-------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Virtual Network (vnet-flynet) | Address space:<br>10.0.0.0/16 | Defines the private IP address range for all Azure resources. Acts as the logical boundary for the Flynet environment, ensuring full network isolation from other tenants.        |
| Subnet snet-web               | Address range:<br>10.0.1.0/24 | Hosts the application tier, including the Azure Load Balancer and Virtual Machine Scale Set. Designed to handle inbound HTTPS requests and outbound connections to data services. |
| Subnet snet-data              | Address range:<br>10.0.2.0/24 | Dedicated to the data tier. Contains private endpoints for Azure Database for PostgreSQL and Blob Storage, preventing any direct public Internet exposure.                        |
| AzureBastionSubnet            | Address range:<br>10.0.3.0/26 | Reserved subnet required for Azure Bastion deployment, enabling secure remote administration of virtual machines through browser-based SSH or RDP without opening external ports. |

Figure 3: Azure Virtual Network (vnet-flynet) with segmented subnets for web, data, and administration layers (Microsoft, 2025).

The virtual network establishes logical separation between Flynet’s cloud environment through subnets that divide the web and data tiers to achieve better security and improved management capabilities. The private backbone of Azure enables internal communication, which decreases both network latency and external security threats. Implementing private endpoints in the data subnet, along with Bastion for administration, aligns with Microsoft's recommended approach to network segmentation and access control (Microsoft, 2025).

## Web Tier Configuration

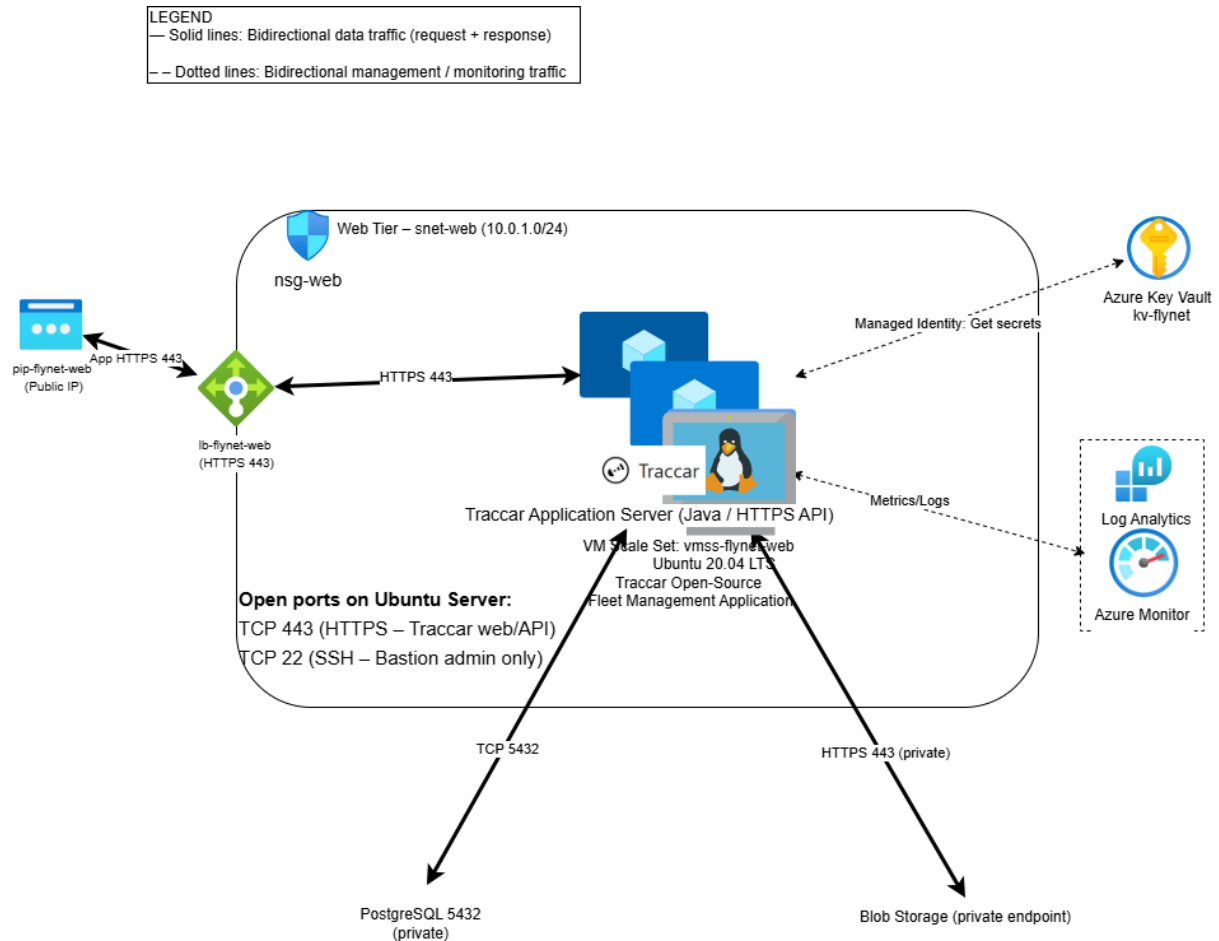


Figure 4: The detailed diagram of the VM Scale Set configuration, perhaps showing instances behind the load balancer.



| Component                                   | Configuration                                                                                                                                                                                                                    | Purpose                                                                                                                                                          |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Public IP (pip-flynet-lb)                   | Standard SKU, Static                                                                                                                                                                                                             | Fixed client access endpoint for dispatcher and driver connections, resolved via Public DNS (flynetfleet.cloudapp.net).                                          |
| Azure Load Balancer                         | Frontend Port: 443 (HTTPS)                                                                                                                                                                                                       | Balances incoming HTTPS requests across VM instances in the scale set.                                                                                           |
| Virtual Machine Scale Set (vmss-flynet-web) | OS: Ubuntu 20.04 LTS<br>Size: Standard_B2s (2 vCPU, 4 GB RAM)<br>Auto-scale range: 2–6 instances (CPU threshold 30–70%)<br>Disk: Premium SSD<br>Open Ports: TCP 443 (HTTPS – Traccar Web/API), TCP 22 (SSH – Bastion admin only) | Hosts the open-source Traccar fleet-management application; provides secure web/API access over HTTPS and administrative SSH access restricted to Azure Bastion. |
| Managed Identity                            | Enabled                                                                                                                                                                                                                          | Allows secure access to Azure Key Vault without stored credentials.                                                                                              |
| DNS Integration                             | Public DNS resolves flynetfleet.cloudapp.net → pip-flynet-lb<br>Private DNS resolves PostgreSQL and Blob endpoints internally                                                                                                    | Ensures correct external and internal name resolution while keeping data traffic within the private network.                                                     |

*Figure 5: The configuration and purpose of the web tier architecture (Microsoft, 2025).*

The web tier receives all incoming traffic on HTTPS at a public endpoint and distributes the load to scale-out VM instances where Traccar is hosted. Managed Identity provides secure access to Azure Key Vault without requiring any password, and DNS integration delivers both public routing (for users) and private name resolution (for internal data connections) (Microsoft, 2025).

Standard\_B2s provides organisations with a balanced CPU–memory ratio, which supports web applications with typical usage needs at budget-friendly prices and enables the organisation to scale operations for higher demand.

## Data Tier Configuration

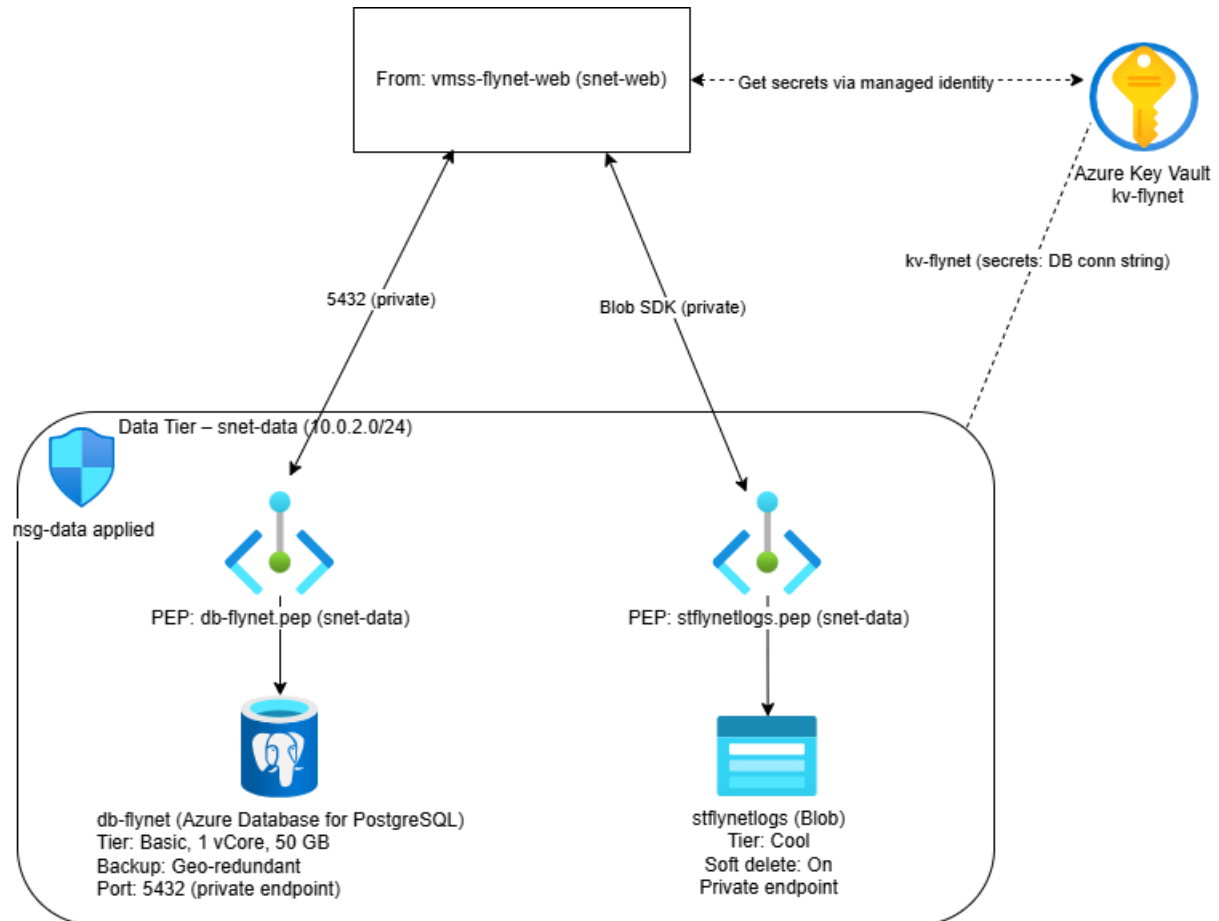


Figure 6: The diagram for the data tier shows the private endpoints and how they connect the subnets.

| Component                                 | Configuration                                                                                             | Purpose                                                                       |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Azure Database for PostgreSQL (db-flynet) | Tier: Basic (1 vCore, 50 GB)<br>Backup: Geo-redundant<br>Network: Private endpoint (pep-pg)<br>Port: 5432 | Stores journey and driver data with managed reliability and automatic backup. |
| Azure Blob Storage (stflynetlogs)         | Tier: Cool<br>Private endpoint: pep-blob<br>Soft delete: Enabled                                          | Stores application logs and archived booking records cost-effectively.        |

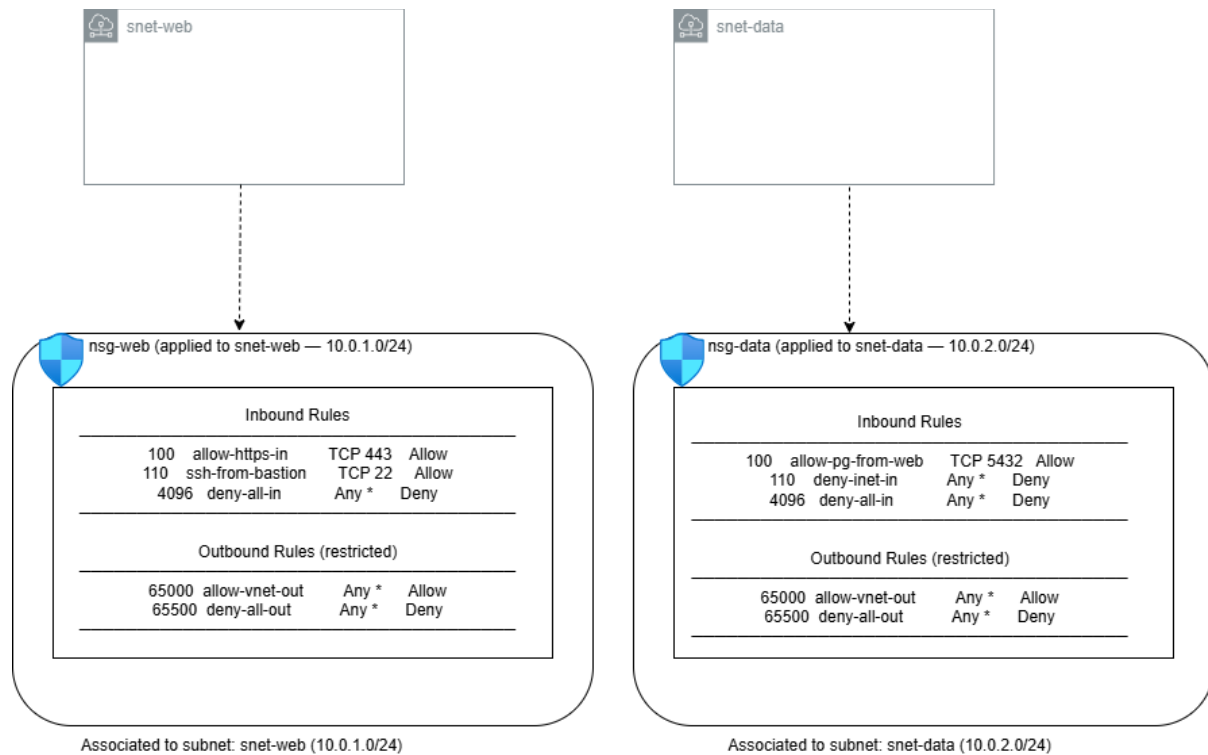
*Figure 7: The configuration and purpose of data tier architecture (Microsoft, 2025).*

The data tier resides in a private subnet, and all the traffic is kept within the Azure network. You may want transactional reliability and geo-redundant PostgreSQL backups, but you can require durable, low-cost storage for logs and historical data with Blob Storage. Both services use private endpoints to avoid exposure to the public Internet (Microsoft, 2025).

The selection of PostgreSQL over Azure SQL Database occurred because the open-source fleet-management application Traccar operates natively on PostgreSQL, which provides both compatibility and simplified configuration and eliminates the need for proprietary database platform licensing expenses.



## Security Configuration



*Figure 8: The Network Security Group (NSG) rules protect web and data subnets through inbound and outbound traffic restrictions, which establish secure communication paths between application tiers.*



| NSG      | Applied To | Rule Summary                                                                      | Purpose                                                                          |
|----------|------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| nsg-web  | snet-web   | Allow HTTPS from Internet; Allow SSH from Bastion; Deny all other inbound traffic | Protects the application tier while permitting secure web and admin access.      |
| nsg-data | snet-data  | Allow PostgreSQL only from snet-web; Deny all inbound external access             | Ensures strict data isolation and prevents unauthorised access to the data tier. |

*Figure 9: NSG configuration and purpose (Microsoft, 2025).*

### Additional Security Controls:

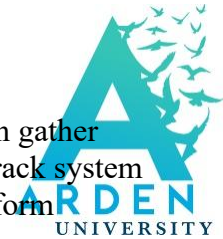
Users can access Azure Bastion through a browser-based SSH connection, which operates through a secure tunnel that does not need public SSH ports. The database connection credentials stored in kv-flynet are accessible through managed identity in Azure Key Vault (kv-flynet). TLS encryption is enforced end-to-end.

Every Network Security Group serves as a virtual firewall, controlling traffic to and from the subnets. Public HTTPS and Bastion-only SSH are administered through the web NSG. The facts NSG restricts database access to only allow internal requests from the web tier. This architecture is in accordance with the Microsoft cloud security model, which involves different layers (Microsoft, 2025).

### Monitoring and Logging

| Service                            | Configuration                            | Purpose                                                                                                  |
|------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Azure Monitor + Log Analytics      | Default diagnostic log ingestion enabled | Tracks performance trends and generates alerts for resource health and performance.                      |
| VMSS Metrics & PostgreSQL Insights | Forwarded to Log Analytics Workspace     | Enables operational troubleshooting and scaling decisions based on system and database performance data. |

*Figure 10: The monitoring architecture configuration and purpose (Microsoft, 2025).*



The monitoring system operates through Azure Monitor and Log Analytics, which gather data from all resources deployed in the system. The tools allow organisations to track system health and scaling operations and detect system faults, which enables them to perform proactive maintenance and achieve operational reliability (Microsoft, 2025).

Configuration Parameters, Interface Settings and Resource Allocations

This section provides the configuration details for each resource deployed in the cloud architecture. The parameters listed below are based on standard Azure deployment settings and reflect the configuration used in the implementation.

Virtual Network and Subnet Configuration

| Component       | Setting       | Value              | Purpose                                                |
|-----------------|---------------|--------------------|--------------------------------------------------------|
| Virtual Network | Name          | vnet-flynet        | Central private network for all application resources. |
|                 | Address Space | 10.0.0.0/16        | Enables subnet segmentation without IP overlap.        |
| Web Subnet      | Name          | snet-web           | Hosts the Virtual Machine Scale Set and Load Balancer. |
|                 | Address Range | 10.0.1.0/24        | Dedicated address space for the application tier.      |
| Data Subnet     | Name          | snet-data          | Hosts PostgreSQL and Blob Storage private endpoints.   |
|                 | Address Range | 10.0.2.0/24        | Isolated subnet that restricts external data access.   |
| Bastion Subnet  | Name          | AzureBastionSubnet | Required subnet name for Azure Bastion service.        |
|                 | Address Range | 10.0.3.0/26        | Minimum supported size for Bastion deployment.         |

Figure 11: Azure Virtual Network configuration and purpose (Microsoft, 2025).

These configurations define Flynet's private network topology, ensuring address segmentation and isolation between tiers. The network architecture is divided into separate subnets for applications, data, and administration, which follow Azure networking standards to enable scalability and secure access (Microsoft, 2025).



The specified address ranges create network separation between tiers, which follows the guidelines for cloud network design as described by Chandrasekaran (2015).

## Public Access and Load Balancer Configuration

| Component     | Setting       | Value                               | Purpose                                                                               |
|---------------|---------------|-------------------------------------|---------------------------------------------------------------------------------------|
| Public IP     | Name          | pip-flynet-lb                       | Provides the public entry point for dispatcher and driver connections.                |
|               | SKU           | Standard                            | Required for secure Load Balancer configuration.                                      |
|               | Allocation    | Static                              | Ensures a fixed, stable endpoint for external access.                                 |
| Load Balancer | Name          | lb-flynet-web                       | Distributes incoming HTTPS requests across VM instances in the scale set.             |
|               | SKU           | Standard, Public                    | Supports load balancing for the auto-scaling backend pool.                            |
|               | Frontend Port | 443 (HTTPS)                         | Handles encrypted client traffic for secure communication.                            |
|               | Backend Pool  | vmss-flynet-web scale set instances | Balances application traffic evenly across virtual machines.                          |
|               | Health Probe  | TCP 80                              | Monitors application health and triggers failover if an instance becomes unavailable. |

Figure 12: Azure Public Access and Load Balancer Configuration (Microsoft, 2025).

The Load Balancer distributes HTTPS traffic through a single static public IP to provide both secure and consistent access. The system monitors VM health by using TCP probes and automatically redirects traffic to other nodes when any node goes down. The configuration design provides system availability, fault tolerance, and scalability as documented in Microsoft's guidance (Microsoft, 2025).



Virtual Machine Scale Set Configuration

| Component         | Setting                             | Value                                                                    | Purpose                                                                 |
|-------------------|-------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------|
| VM Scale Set      | Name                                | vmss-flynet-web                                                          | Hosts the open-source Traccar fleet-management application.             |
|                   | OS Image                            | Ubuntu Server 20.04 LTS (Canonical)                                      | Verified, stable Linux distribution optimised for cloud workloads.      |
|                   | VM Size                             | Standard_B2s (2 vCPU, 4 GB RAM)                                          | Provides a cost-effective compute tier for moderate usage.              |
|                   | Instance Count                      | Minimum 2, Maximum 6                                                     | Supports automatic scaling based on real-time demand.                   |
|                   | Auto-scale Rule                     | Scale-out $\geq$ 70% CPU · Scale-in $\leq$ 30% CPU                       | Dynamically adjusts capacity to maintain performance and minimise cost. |
|                   | Disk                                | Premium SSD, 32 GB                                                       | Offers high IOPS and low latency for consistent performance.            |
| Managed Identity  | Enabled                             | Grants secure access to Azure Key Vault without embedding credentials.   |                                                                         |
| Outbound Internet | Disabled (except via Load Balancer) | Prevents accidental data exposure and enforces outbound traffic control. |                                                                         |

Figure 13: Azure Virtual Machine Scale Set configuration (Microsoft, 2025).

The Virtual Machine Scale Set (VMSS) serves as the main computing infrastructure for Flynet’s web tier. It runs the Traccar application across multiple instances for load distribution and failover. Microsoft (2025) states that auto-scaling policies with managed identity integration and restricted outbound access provide both elasticity and security and compliance.

The configuration provides both performance and cost efficiency, and secure credential management.

## Data Tier Configuration

| Component                                     | Setting          | Value                                  | Purpose                                                            |
|-----------------------------------------------|------------------|----------------------------------------|--------------------------------------------------------------------|
| Azure Database for PostgreSQL Flexible Server | Name             | db-flynet                              | Stores booking, driver, and journey data for operational tracking. |
|                                               | Compute Tier     | Basic – 1 vCore                        | Optimised for small fleet workloads with moderate concurrency.     |
|                                               | Storage          | 50 GB                                  | Allocated based on projected growth of trip and user records.      |
|                                               | Backup Retention | 7 days (Geo-redundant)                 | Provides automatic recovery in case of failure or data corruption. |
|                                               | Network Access   | Private Endpoint (pep-pg) in snet-data | Restricts access to internal network; no public exposure.          |
|                                               | Port             | 5432 (private)                         | Standard PostgreSQL port for database communication.               |
| Blob Storage Account                          | Name             | stflynetlogs                           | Stores logs, archived trip receipts, and exported data.            |
|                                               | Tier             | Cool storage tier                      | Reduces cost for infrequently accessed historical data.            |
|                                               | Private Endpoint | pep-blob (in snet-data)                | Enables secure internal connectivity only.                         |
|                                               | Data Protection  | Soft delete enabled                    | Safeguards data from accidental deletion.                          |

Figure 14: Azure Data Tier configuration (Microsoft, 2025).

The PostgreSQL Flexible Server maintains automatic reliability and backup systems and high availability features, but Blob Storage offers scalable and cost-effective storage for log and record retention. The data subnet includes Private Endpoints, which create separate networks for both services while implementing strict access rules and following Azure security standards (Microsoft, 2025).

## Network Security Groups (NSG) Configuration

| NSG      | Rule Name               | Priority | Source → Destination                        | Port / Protocol | Action | Purpose                                                            |
|----------|-------------------------|----------|---------------------------------------------|-----------------|--------|--------------------------------------------------------------------|
| nsg-web  | allow-https-in          | 100      | Internet → <u>snet-web</u>                  | TCP 443         | Allow  | Enables secure HTTPS access for dispatcher and driver connections. |
|          | allow-ssh-from-bastion  | 110      | <u>AzureBastionSubnet</u> → <u>snet-web</u> | TCP 22          | Allow  | Permits administrative SSH access only through Azure Bastion.      |
|          | deny-all-in             | 4096     | Any → <u>snet-web</u>                       | Any             | Deny   | Blocks all other inbound traffic to reduce the attack surface.     |
| nsg-data | allow-postgres-from-web | 100      | <u>snet-web</u> → <u>snet-data</u>          | TCP 5432        | Allow  | Authorises PostgreSQL traffic only from the application subnet.    |
|          | deny-internet-in        | 110      | Internet → <u>snet-data</u>                 | Any             | Deny   | Prevents any direct external access to the data tier.              |
|          | deny-all-in             | 4096     | Any → <u>snet-data</u>                      | Any             | Deny   | Enforces strict isolation and least-privilege communication.       |

Figure 15: Network Security Groups (NSG) configuration (Microsoft, 2025).

The NSG operates as a virtual firewall system which controls all network traffic that enters or leaves a subnet. The web tier allows only HTTPS and Bastion-admin traffic, while the data tier restricts all access except PostgreSQL from the web subnet. The multi-layered security system within Azure Zero Trust network architecture prevents attackers from moving between VNet resources (Microsoft, 2025).

## Identity, Secrets, Logging, and Monitoring

| Component         | Setting        | Value                                                                         | Purpose                                                                                                                                             |
|-------------------|----------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Azure Key Vault   | Name           | kv-flynet                                                                     | Stores database connection strings and credentials securely.                                                                                        |
|                   | Access Method  | Managed Identity (no passwords in code)                                       | Prevents credential leakage by allowing the VM Scale Set to retrieve secrets directly through Azure Active Directory.                               |
| Public DNS        | Domain Mapping | flynetfleet.cloudapp.net → pip-flynet-lb                                      | Resolves the external service name for dispatcher and driver access through HTTPS.                                                                  |
| Private DNS Zones | Internal Names | privatelink.postgres.database.azure.com;<br>privatelink.blob.core.windows.net | Provides internal name resolution for PostgreSQL and Blob Storage <b>private endpoints</b> , ensuring all traffic remains within the Azure network. |
| Azure Bastion     | Name           | bastion-flynet                                                                | Enables secure browser-based SSH/RDP access for administrators without exposing public ports.                                                       |
| Monitoring        | Services       | Azure Monitor + Log Analytics                                                 | Collects performance metrics, system logs, and alerts for observability and fault detection across all tiers.                                       |

Figure 16: Identity, secrets, logging, and monitoring configuration (Microsoft, 2025).

Identity and monitoring controls protect operational integrity by enforcing compliance standards for organisations. Azure Key Vault protects credentials, and Managed Identity functions as a static secret replacement, while Private DNS Zones enable private endpoint resolution within the internal network. Azure Monitor and Log Analytics provide organisations with a unified observability solution that enables them to monitor their systems through one interface, which helps them identify problems before they escalate into major issues (Microsoft, 2025).

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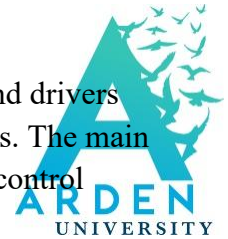
## System Interaction and Data Flow

The deployed infrastructure is organised as a two-tier architecture, separating the web application layer from the data storage layer. All external access enters the system through a single public endpoint, and internal communication is restricted to private network channels within the virtual network. The system design implements a suitable framework for fleet management applications which require protected remote access and reliable request handling and stable data storage. The layered cloud application design structure provides better scalability and fault isolation, according to Chandrasekaran (2015).



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The system infrastructure enables protected data exchange between dispatchers and drivers through its design yet it limits administrative access to particular permission levels. The main functions of each tier appear in the following tables which include their essential control systems.



## Client Access

| User Type   | Access Method                | Connection Path                                                            | Security / Rationale                                                  |
|-------------|------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Dispatchers | Web browser<br>(home/remote) | flynetfleet.cloudapp.net → DNS<br>resolves to pip-flynet-lb → HTTPS<br>443 | Central DNS entry simplifies control and<br>enables encrypted access. |
| Drivers     | Mobile app                   | Same HTTPS endpoint (443) via<br>Public IP                                 | Ensures consistent routing and secure<br>mobile communication.        |

Figure 17: User access and entry flow

The system has one HTTPS endpoint for client authentication that allows unified logging and access inspection.

## Application Processing Layer (Web Tier)

| Component                                      | Configuration                                 | Function                                                                       | Justification                                                       |
|------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Azure Load Balancer (lb-flynet-web)            | Standard tier, public IP<br>association       | Distributes incoming HTTPS<br>traffic across VMSS instances                    | Provides redundancy and even<br>traffic distribution.               |
| Virtual Machine Scale Set<br>(vmss-flynet-web) | Ubuntu 20.04 LTS, auto-scale<br>2–6 instances | Hosts <b>Traccar</b> app: manages<br>bookings, GPS, and driver<br>availability | Enables automatic scaling and<br>elasticity (Chandrasekaran, 2015). |

Figure 18: Application layer configuration

## Data Storage Layer (Data Tier)

| Resource                    | Access Method                     | Subnet                | Security / Protocol           | Purpose                                 |
|-----------------------------|-----------------------------------|-----------------------|-------------------------------|-----------------------------------------|
| PostgreSQL (db-flynet)      | Private Endpoint (pep-pg)         | snet-data             | TCP 5432 (only from snet-web) | Persistent operational data storage     |
| Blob Storage (stflynetlogs) | Private Endpoint (pep-blob)       | snet-data             | Internal network only         | Stores logs and archived booking data   |
| Private DNS Zones           | privatelink.postgres... / blob... | Linked to vnet-flynet | Internal resolution only      | Keeps data traffic inside Azure network |

Figure 19: Data and storage configuration

The system design achieves least privilege access control by eliminating all public database exposure which meets cloud-security standards (Jansen & Grance, 2011).

## Identity and Credential Security

| Component                   | Function                            | Security Advantage                           |
|-----------------------------|-------------------------------------|----------------------------------------------|
| Azure Key Vault (kv-flynet) | Stores PostgreSQL connection string | Secrets encrypted and centrally managed      |
| Managed Identity            | Authenticates VMSS to Key Vault     | Prevents credentials in config files or code |

Figure 20: Identity and secret management

The method follows NIST standards because it removes fixed secrets and enables credential renewal without requiring application redeployment (Jansen & Grance, 2011).

## Monitoring and Administration

| Tool / Service                          | Function                           | Benefit                                                    |
|-----------------------------------------|------------------------------------|------------------------------------------------------------|
| Azure Monitor + Log Analytics Workspace | Collects metrics, logs, and alerts | Centralised monitoring for reliability and fault detection |
| Azure Bastion                           | Browser-based SSH access to VMSS   | Administrative access without public SSH exposure          |

Figure 21: Observability and administration controls

Azure Monitor enables proactive maintenance operations and trend analysis, and Bastion provides secure operations through its portless architecture.

## Summary of Project Flow

| Step | Communication                                                            | Purpose                                                                           |
|------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1    | Dispatcher/Driver → <b>Public DNS</b> → <b>pip-flynet-lb (HTTPS 443)</b> | External name resolution and secure system entry point.                           |
| 2    | Load Balancer ↔ VMSS (in snet-web)                                       | Request distribution and response handling between users and application servers. |
| 3    | VMSS ↔ PostgreSQL via pep-pg (TCP 5432)                                  | Booking, driver, and trip data storage through the private database endpoint.     |
| 4    | VMSS ↔ Blob Storage via pep-blob (private HTTPS 443)                     | Log and record archival within the private network.                               |
| 5    | VMSS ↔ Azure Key Vault (Managed Identity)                                | Secure credential retrieval without hard-coded secrets.                           |
| 6    | VMSS ↔ Private DNS Zones                                                 | Internal endpoint name resolution for PostgreSQL and Blob Storage.                |
| 7    | Resource metrics ↔ Azure Monitor / Log Analytics                         | Observability, fault detection, and performance monitoring.                       |
| 8    | Admin ↔ Azure Bastion ↔ VMSS                                             | Secure browser-based administrative access without public SSH exposure.           |

Figure 22: The overview table of Project Flow (Microsoft, 2025).



This bidirectional data flow in the table shows how requests travel securely through the Azure environment. The Public DNS enables outside HTTPS connections, but all internal data exchange occurs through private endpoints and DNS zones. The system follows Azure's least-privilege principle for each connection to ensure compliance with the Zero-Trust architecture (Microsoft, 2025).

The system design ensures dispatchers have secure access from any location, maintains operational stability during peak periods, and safeguards critical operational data, adhering to cloud security and scalability principles outlined in the module.

## Task 2 Report

### Cloud Service Model and Deployment Model

The proposed solution combines Infrastructure-as-a-Service (IaaS) with Platform-as-a-Service (PaaS) capabilities, which are available through Microsoft Azure's Public Cloud platform. This design removes the need for local hardware, supports remote dispatching, and allows flexible scaling (Laudon and Laudon, 2022).

| Layer               | Azure Service Used                          | Role                                       | Justification                                                      | Model Type   |
|---------------------|---------------------------------------------|--------------------------------------------|--------------------------------------------------------------------|--------------|
| Application tier    | Virtual Machine Scale Set (vmss-flynet-web) | Hosts Traccar open-source server           | Requires full OS-level control and custom runtime configuration    | IaaS         |
| Data tier           | Azure Database for PostgreSQL               | Stores fleet, booking, and driver data     | PaaS handles patching, backup, and recovery (Chandrasekaran, 2015) | PaaS         |
| Storage             | Azure Blob Storage (Cool tier)              | Retains logs and receipts                  | Managed redundancy and lifecycle management                        | PaaS         |
| Overall environment | Microsoft Azure Cloud (UK South)            | Shared, internet-accessible infrastructure | No local datacentre; minimal capital expense                       | Public Cloud |

*Figure 23: Service and Deployment Models*

Flynet operates the application layer through this IaaS and PaaS hybrid model, which allows Azure to handle infrastructure reliability and maintenance tasks. IaaS allows users to configure Traccar's Linux runtime, but PaaS provides managed backup and high availability for persistent data.

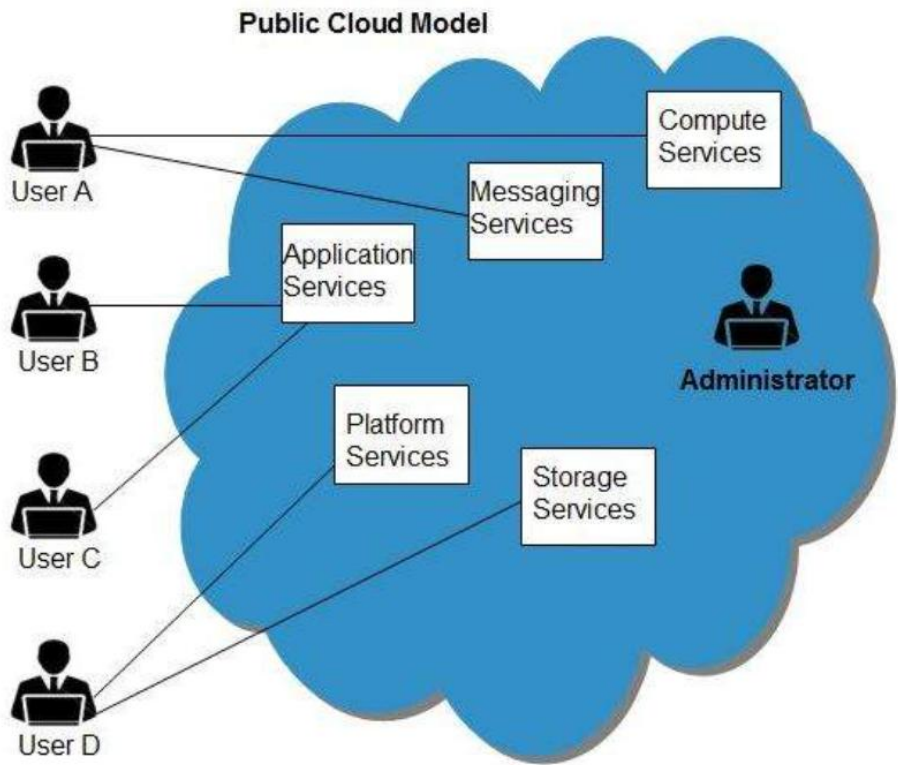


Figure 24: Public Cloud model (Tutorialspoint, 2025).

| Responsibility Area                    | Managed by Azure | Managed by Flynet |
|----------------------------------------|------------------|-------------------|
| Physical datacentre, network, hardware | ✓                | –                 |
| Virtualisation, OS patching (for PaaS) | ✓                | –                 |
| OS patching, runtime config (for IaaS) | –                | ✓                 |
| Application, data, and user access     | –                | ✓                 |

Figure 25: Shared Responsibility in Azure

The shared-responsibility model defines Azure as the platform security handler, but Flynet retains full authority to manage application-level configurations and access permissions. Microsoft (2025) states that these systems work together to help organizations achieve better scalability and security and improved operational performance.

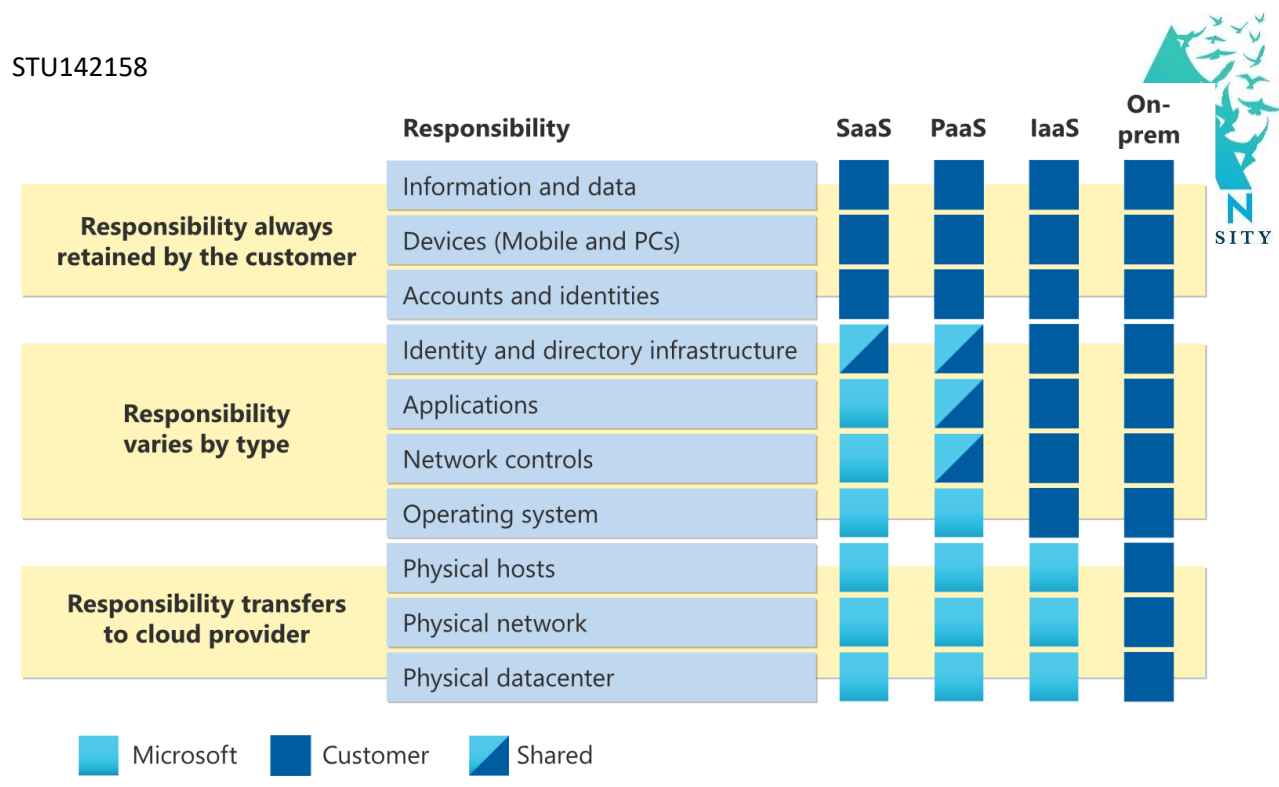


Figure 26: Shared Responsibility Model in Microsoft Azure for SaaS, PaaS and IaaS (Microsoft, 2025).

## Justification of Configuration and Design Choices

The infrastructure is designed as a two-tier distributed architecture, separating the application tier from the data tier. The system benefits from scalability, maintainability, and fault isolation because failures in one layer cannot spread throughout the system (Chandrasekaran, 2015). The tables below summarise the main design choices, configuration parameters and rejected alternatives.

### Application Platform (Traccar)

| Aspect                    | Selected option                    | Justification                                                                                                        | Alternatives rejected                                            |
|---------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Fleet-management software | Traccar (open-source)              | Supports GPS tracking, dispatcher coordination and relational database storage at zero licence cost (Traccar, 2025). | Commercial SaaS fleet tools – recurring fees and vendor lock-in. |
| Licensing model           | Open-source (self-hosted)          | Suitable for a small private company with limited budget; full control over deployment and data.                     | Proprietary licences with per-vehicle or per-user fees.          |
| Data model                | Relational (PostgreSQL-compatible) | Aligns with cloud PaaS relational services and supports transactional consistency for bookings.                      | NoSQL stores, which would add mapping complexity.                |

Figure 27: Justification of application platform choice

The system meets Flynet's operational requirements through its cloud-based relational database design which removes future licensing expenses (Traccar, 2025; Valacich, George and Hoffer, 2023).

## Virtual Network and Subnet Segmentation

| Component          | Configuration | Role / Justification                                                                                       | Alternative rejected                               |
|--------------------|---------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| vnet-flynet        | 10.0.0.0/16   | Single private address space that can be segmented into tiers and extended as Flynet grows.                | Multiple disjoint VNets – adds early complexity.   |
| snet-web           | 10.0.1.0/24   | Isolates the application tier (VMSS and Load Balancer). Limits exposure to only required internet traffic. | Flat network – weaker access control.              |
| snet-data          | 10.0.2.0/24   | Hosts PostgreSQL and Blob Storage private endpoints; no direct internet route, following least privilege.  | Data in web subnet – higher lateral-movement risk. |
| AzureBastionSubnet | 10.0.3.0/26   | Dedicated subnet required by Azure Bastion for secure admin access without public SSH.                     | Public SSH on VMs – common attack vector.          |

Figure 28: Virtual network and subnet design

The practice of subnet segmentation creates isolated networks, which helps prevent attackers from moving between systems after gaining access to one network segment, according to distributed system security recommendations from Jansen and Grance (2011) and Chandrasekaran (2015).

## Ingress and Load Balancing

| Component                      | Use in design                                         | Reason for selection                                                | Alternative / reason rejected                                                                                        |
|--------------------------------|-------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| pip-flynet-lb (Public IP)      | Single HTTPS entry point for dispatchers and drivers. | Stable DNS mapping, simplifies firewall rules and monitoring.       | Dynamic IP – would complicate DNS and client configuration.                                                          |
| Azure Load Balancer (Standard) | Distributes HTTPS traffic across VMSS instances.      | Provides high availability and basic Layer 4 balancing at low cost. | Azure Application Gateway – adds WAF and L7 routing, but unnecessary and more expensive for internal-style use only. |

Figure 29: Ingress and load balancing choices



A Standard Load Balancer fulfils Flynet's present requirements for availability and simplicity, yet provides a future upgrade option to Application Gateway with WAF when the company needs to implement a public customer portal.

### Compute Tier – VM Scale Set (vmss-flynet-web)

| Parameter         | Value / Choice                                      | Justification                                                                                                          | Alternative rejected                                                    |
|-------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Compute model     | Azure Virtual Machine Scale Set (VMSS)              | Supports horizontal scaling and automatic instance replication, key for handling variable load (Chandrasekaran, 2015). | Single VM – no resilience or elastic scaling.                           |
| OS image          | Ubuntu Server 20.04 LTS                             | Stable, long-term support Linux distribution widely used for Traccar deployments.                                      | Windows Server – higher licensing cost, not required for this workload. |
| VM size           | Standard_B2s (2 vCPU, 4GB RAM)                      | Balanced CPU/RAM profile suitable for a medium-load web application, while keeping cost low.                           | Larger sizes (e.g. D-series) – unnecessary at current scale.            |
| Instance range    | Min 2, Max 6                                        | Two instances provide baseline redundancy; up to six cover peak periods without over-provisioning.                     | Fixed single instance – no failover or load handling.                   |
| Auto-scale rules  | Scale-out at $\geq 70\%$ CPU; in at $\leq 30\%$ CPU | Adjusts capacity automatically to demand, reducing manual intervention and cost.                                       | Static capacity – either wasteful or fragile.                           |
| Deployment target | Traccar application server                          | VMSS allows full control over ports and services required by Traccar.                                                  | Azure App Service – limited control over system-level configuration.    |

*Figure 30: Justification of compute configuration*

The use of VMSS provides Flynet with control and elasticity and resilience which aligns with the distributed system scalability and fault tolerance learning outcomes of the module (Valacich, George and Hoffer, 2023).

## Database and Storage Layer

| Component                                 | Configuration / Role                                   | Justification                                                                                                | Alternative rejected                                                                   |
|-------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Azure Database for PostgreSQL (db-flynet) | Basic tier, 1 vCore, 50GB, private endpoint (pep-pg)   | Native Traccar support, relational consistency for bookings, automated patching and backups.                 | Azure SQL Database – would require app changes and add proprietary licensing overhead. |
| Network access to DB                      | Private Endpoint in snet-data                          | No public internet exposure; only VMSS instances in snet-web can connect (least-privilege model).            | Public endpoint with firewall only – larger attack surface.                            |
| Azure Blob Storage (stflynetlogs)         | Cool tier, private endpoint (pep-blob), soft delete on | Cost-effective for infrequently accessed logs and receipts; soft delete protects against accidental removal. | Local VM disk for logs – harder to retain, scale and secure.                           |

Figure 31: Data and storage choices

The configuration protects operational data and logs through privacy measures and cost efficiency while keeping full compatibility with the selected application stack (Traccar, 2025).

## Network Security and Administrative Access

| Control       | Key rules / use                                                 | Justification                                                                                                                     | Alternative rejected                                                          |
|---------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| nsg-web       | Allow HTTPS from internet; SSH only from Bastion; deny all else | Restricts inbound traffic to the minimum needed for the web tier.                                                                 | Open SSH from internet – high risk of brute-force attacks.                    |
| nsg-data      | Allow PostgreSQL (5432) only from snet-web; deny internet       | Ensures the data tier is reachable only by the application tier, preventing direct external access.                               | Allow internet to DB – violates least privilege and increases attack surface. |
| Azure Bastion | Browser-based SSH/RDP via Azure portal                          | Provides secure admin access without exposing port 22/3389 publicly, aligning with cloud best practice (Jansen and Grance, 2011). | Public SSH on each VM – common source of compromise.                          |

Figure 32: Security control justification

For Flynet's current scale, NSGs and Bastion provide sufficient protection without the cost and added routing complexity of Azure Firewall. A firewall should be added later when the network grows to include multiple regions or VNets.

## Credential and Name Resolution Management

| Component                  | Role in design                                                                | Justification                                                                                                                                                       | Alternative rejected                                                                              |
|----------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Azure Key Vault            | Stores PostgreSQL connection strings and sensitive config.                    | Centralises secret management; enables secure retrieval and rotation without embedding passwords in code or VM disks (Jansen and Grance, 2011).                     | Plain text config files or environment variables on VMs – exposed if images or logs are accessed. |
| Private DNS Zones          | privatelink.postgres.database.azure.com,<br>privatelink.blob.core.windows.net | Provide internal name resolution for private endpoints so VMSS instances can always reach DB and Storage through private IPs, even as they scale (Microsoft, 2024). | Manual host file entries on VMs – error-prone and not scalable with auto-scaling.                 |
| Public DNS (Azure default) | Resolves the public IP (pip-flynet-lb) for HTTPS access.                      | Allows dispatcher and driver devices to reach the application securely over the internet without exposing backend services.                                         | Direct IP configuration on clients – fragile if IP ever changes and harder to manage.             |

*Figure 33: Secrets and DNS strategy*

The combination of Public DNS for frontend operations with Private DNS for backend services enables proper name resolution between tiers while maintaining network isolation and data security (Chandrasekaran, 2015).

## Scalability and Reliability of the Proposed Infrastructure

The system architecture always operates while maintaining operational performance through various operational needs and continues to function even when any system component fails. Scalability is primarily achieved through the Azure Virtual Machine Scale Set (VMSS) deployed in the web tier. The VMSS system uses CPU-utilisation thresholds to determine when it should scale up or down the number of application instances. The system uses more instances during peak travel hours to process many booking requests simultaneously but reduces instance numbers when demand is low to reduce expenses. The system operates based on distributed system horizontal scaling principles because it expands its capacity by adding new nodes (Chandrasekaran, 2015).

Scale out during each work day

Scale mode
☐ Scale based on a metric
☒ Scale to a specific instance count

Instance count\* ⓘ
 ✓

Schedule
☐ Specify start/end dates
☒ Repeat specific days

Repeat every
☒ Monday
☒ Tuesday
☒ Wednesday
☒ Thursday
☒ Friday
☐ Saturday
☐ Sunday

Timezone
 ▼

Start time

End time

Specify an end time, else this scale condition will apply for all days until it reaches the start time of another scale condition

Scale in during evening

Scale mode
☐ Scale based on a metric
☒ Scale to a specific instance count

Instance count\* ⓘ
 ✓

Schedule
☐ Specify start/end dates
☒ Repeat specific days

Repeat every
☒ Monday
☒ Tuesday
☒ Wednesday
☒ Thursday
☒ Friday
☐ Saturday
☐ Sunday

Timezone
 ▼

Start time
 ✓

End time
 ✓

Specify an end time, else this scale condition will apply for all days until it reaches the start time of another scale condition

Figure 34: Autoscale configuration in Azure portal (Microsoft, 2025).

The Azure Load Balancer distributes incoming HTTPS traffic among active VM instances to improve both performance and reliability. The system distributes workloads across multiple virtual machines to prevent any single machine from reaching capacity while maintaining service continuity during instance failures. The system functions as a fault-tolerant system because it continues to operate correctly when any individual component stops working (Valacich, George, and Hoffer, 2023).

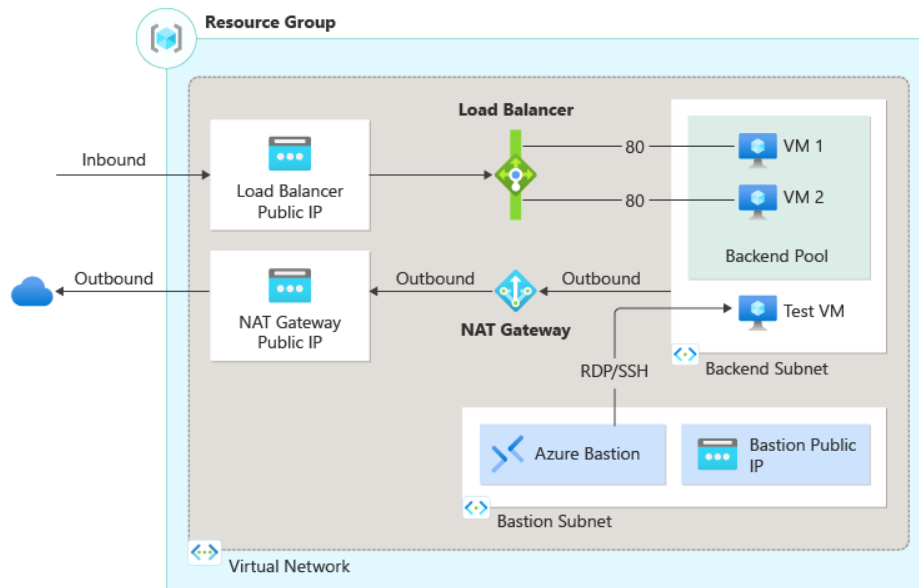


Figure 35: Azure Load Balancer – backend pool configuration (Microsoft, 2025).

Reliability in the data layer is strengthened using Azure Database for PostgreSQL Flexible Server, which provides automated patching, monitoring, backup scheduling and fault recovery. Geo-redundant backups are enabled, meaning data can be restored to an alternative region in the event of a regional failure. Azure Blob Storage protects archived data and system logs through Microsoft-managed redundancy which eliminates the need for manual replication according to Jansen and Grance (2011).

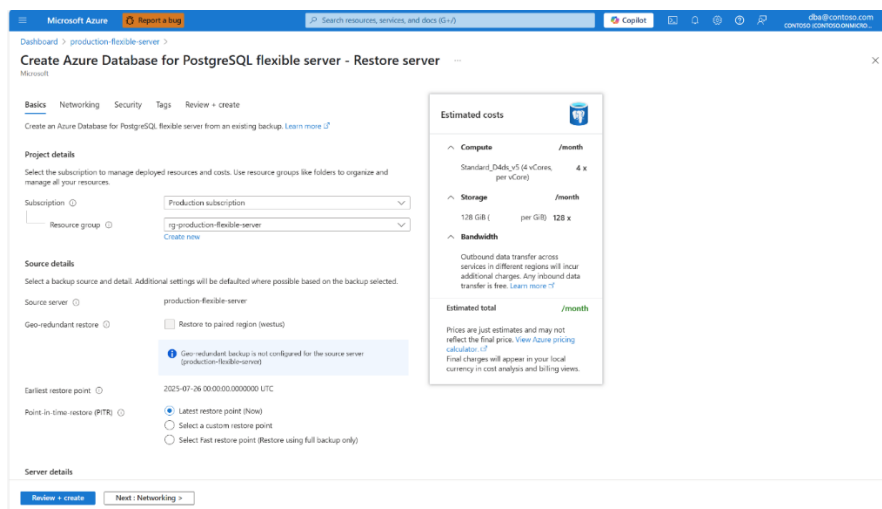


Figure 36: Azure Database for PostgreSQL Flexible Server – geo-redundant backup configuration (Microsoft, 2025).

Private Endpoints and Network Security Groups (NSGs) operate as a system to enhance network reliability and security. The database and storage services are not publicly addressable; only the VMSS in the web subnet can communicate with them. The system becomes more secure because unauthorized access becomes impossible and external disruptions result in less system downtime. The VMSS obtains administrative access through Azure Bastion which allows secure maintenance operations from anywhere without exposing SSH ports to the internet.

The system provides high availability and scalable performance, and controlled, predictable operation, which are necessary for real-time fleet-dispatch systems that need to coordinate taxi services efficiently and reliably.

## Cost Estimate

To validate the financial viability of the proposed Azure deployment, service costs were calculated using the Microsoft Azure Pricing Calculator for the UK South region on 10 November 2025.

The estimate includes all required services for the Flynet fleet-management infrastructure (web tier, data tier, and monitoring). Prices are shown in USD because the calculator defaults to U.S. currency; conversion to GBP would depend on exchange rates at the time of billing.

The total estimated monthly cost is around \$397 USD, which is approximately £317 GBP. The estimated upfront cost is \$0.00 USD (Microsoft, 2025).

The Flynet Transportation cloud solution operates at a monthly budget of £300–£320 based on the scale-set usage and database activity levels. The system operates on a usage-based pricing structure, which enables Flynet to minimise fixed capital costs while maintaining system flexibility and dependability.

| Your Estimate                 |   |                                                              |       |                   |                     |  |  |
|-------------------------------|---|--------------------------------------------------------------|-------|-------------------|---------------------|--|--|
| Virtual Machine Scale Sets    | ① | 1 D1 (1 vCPU, 3.5 GB RAM) x 730 Hours (Pay as you go), ...   | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$102.20 |  |  |
| Virtual Network               | ① | East US (Virtual Network 1): 100 GB Outbound Data Tran...    | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$4.00   |  |  |
| Load Balancer                 | ① | Basic Load Balancer is free of charge                        | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$0.00   |  |  |
| Azure DNS                     | ① | Zone 1, DNS, Public; 0 hosted DNS zones, 0 DNS queries       | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$0.00   |  |  |
| IP Addresses                  | ① | Basic (Classic), 0 Dynamic IP Addresses X 730 Hours, 0 St... | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$0.00   |  |  |
| Azure Bastion                 | ① | Basic Tier, 730 Hours, 5 GB Outbound Data Transfer           | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$138.70 |  |  |
| Storage Accounts              | ① | Block Blob Storage, General Purpose V2, Flat Namespac...     | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$21.84  |  |  |
| Azure Database for PostgreSQL | ① | Flexible Server Deployment, General Purpose Tier, 1 D2d...   | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$130.52 |  |  |
| Key Vault                     | ① | Vault: 0 operations, 0 advanced operations, 0 renewals, ...  | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$0.00   |  |  |
| Azure Monitor                 | ① | Log analytics: Log Data Ingestion: 0 GB Daily Auxiliary L... | ⚙️ 🗑️ | Upfront: US\$0.00 | Monthly: US\$0.00   |  |  |

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#### Support

SUPPORT:

Basic (Included) ⓘ

US\$0.00

#### Select your program/offer

LICENSING PROGRAMME:

Microsoft Customer Agreement (MCA) ⓘ

[Log in](#) to see your Azure agreement pricing.

☒ Show Dev/Test Pricing ⓘ

Estimated upfront cost

US\$0.00

Estimated monthly cost

US\$397.26



*Figure 37: Azure Pricing Calculator – estimated monthly cost for Flynet fleet-management deployment (UK South region) (Microsoft, 2025).*

The tracking of ongoing expenses will be managed through Azure Cost Management + Billing which enables Flynet to monitor expenses through dashboards and forecasts and budget alerts for maintaining controlled monthly operating costs and avoiding resource overuse (Microsoft, 2025).

---

## Maintenance and Monitoring Strategy

The Flynet cloud infrastructure requires ongoing maintenance and performance monitoring to achieve long-term stability and security, and to reduce operational costs. Azure delivers automated tools which help organisations reduce manual work while upholding regulatory standards and operational stability.



| Aspect                                      | Configuration / Actions                                                                                                                                                                                                                                                                                                                                                             | Purpose                                                                                                                                                        |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monitoring and Alerts</b>                | <p>Azure Monitor and Log Analytics collect metrics and diagnostic logs from VMSS, PostgreSQL, and network components.</p> <ul style="list-style-type: none"> <li>• Metric alerts for high CPU, abnormal memory, or failed autoscaling.</li> <li>• Log alerts for PostgreSQL errors or authentication failures.</li> <li>• Dashboards summarising latency and throughput.</li> </ul> | Provides real-time visibility and early-warning notifications, ensuring proactive issue resolution and consistent service reliability (Jansen & Grance, 2011). |
| <b>Patch and Backup Management</b>          | <ul style="list-style-type: none"> <li>• Azure Database for PostgreSQL applies automatic security patches and minor version updates.</li> <li>• Geo-redundant backups retained 7–35 days for disaster recovery.</li> <li>• VMSS automatic OS image upgrades for the web tier.</li> <li>• System logs exported periodically to Blob Storage.</li> </ul>                              | Maintains data integrity, supports recovery from regional failures, and ensures continuous compliance without manual intervention.                             |
| <b>Access Control and Secret Management</b> | <ul style="list-style-type: none"> <li>• Administrative access only via Azure Bastion.</li> <li>• No public SSH/RDP ports exposed.</li> <li>• Secrets and connection strings stored in Azure Key Vault, retrieved via Managed Identity.</li> </ul>                                                                                                                                  | Reduces attack surface and enforces secure, role-based access control for administrators and application services.                                             |
| <b>Cost and Performance Optimisation</b>    | <ul style="list-style-type: none"> <li>• Monthly usage reviewed in Azure Cost Management + Billing.</li> <li>• Autoscaling thresholds tuned to adjust VMSS capacity during off-peak hours.</li> <li>• Alerts set for spending anomalies.</li> </ul>                                                                                                                                 | Ensures predictable operational expenditure and performance tuning aligned with Flynet's usage patterns.                                                       |

Figure 38: Example Azure Monitor and Log Analytics workflow (Microsoft, 2025).



## Recommendations for Final Implementation

The implemented architecture establishes a protected and expandable system which supports Flynet's present operational requirements. The organisation needs to implement additional security measures when it expands its operations into new service areas or grows for better resilience, security, and long-term cost efficiency. The recommendations follow a structured order, which allows Flynet to deploy new features according to its business growth requirements.

| Focus Area                          | Recommendation                                                                                                                                                                                                                                                                                                                                | Purpose / Benefit                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1. Secret Management                | Implement <b>Managed Identity</b> for the Virtual Machine Scale Set (VMSS) to access Azure Key Vault directly through Azure Active Directory.                                                                                                                                                                                                 | Removes stored credentials, enhances security, and simplifies secret rotation (Jansen & Grance, 2011).          |
| 2. Disaster Recovery (Multi-Region) | Adopt a <b>phased strategy</b> for regional failover and data protection:<br><b>Stage 1 (Now):</b> Continue using geo-redundant PostgreSQL backups.<br><b>Stage 2 (Future):</b> Enable PostgreSQL read replica in UK West.<br><b>Stage 3 (Expansion):</b> Deploy standby VMSS + Load Balancer in UK West and configure Azure Traffic Manager. | Improves resilience and reduces recovery time (from hours → minutes) while keeping cost proportional to growth. |
| 3. Network Security                 | Add <b>Azure Firewall</b> once Flynet expands to multiple regions or connects external partners.                                                                                                                                                                                                                                              | Centralises security policies, packet inspection, and outbound filtering beyond NSG scope.                      |
| 4. Compute Optimisation (Future)    | Evaluate:<br><ul style="list-style-type: none"> <li>• <b>Azure Kubernetes Service (AKS)</b> for containerised workloads.</li> <li>• <b>Azure Functions</b> for background or scheduled jobs.</li> </ul>                                                                                                                                       | Supports greater scalability and automation as system demand increases.                                         |
| 5. Cost Control                     | Continue monitoring with <b>Azure Cost Management + Billing</b> dashboards and budget alerts.                                                                                                                                                                                                                                                 | Maintains predictable monthly spend and identifies under-used resources for optimisation.                       |

Figure 39: Azure multi-region business continuity and disaster recovery model (Microsoft, 2025).

## Conclusion:

The project shows how a properly designed cloud infrastructure enables Flynet Transportation to shift from manual coordination to a data-based dispatch system. The proposed design uses Microsoft Azure public cloud to integrate Infrastructure-as-a-Service with Platform-as-a-Service which creates a secure scalable cost-effective base that fulfills present and future business requirements. The separate tiers of application and data in private networking create better reliability and resilience, while VM Scale Sets and PostgreSQL and Blob Storage from Azure maintain performance stability through changing demand patterns.

The solution creates an operational system which helps organisations meet their present targets and establishes a base for future growth. The solution allows Flynet to expand its operations across different regions while enhancing data security and enabling the implementation of future capabilities including managed identities and zone-redundant databases and containerized deployment. The architecture behind Flynet enables its current operational efficiency through performance and security features which also provide flexibility for future technological and organizational developments.

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